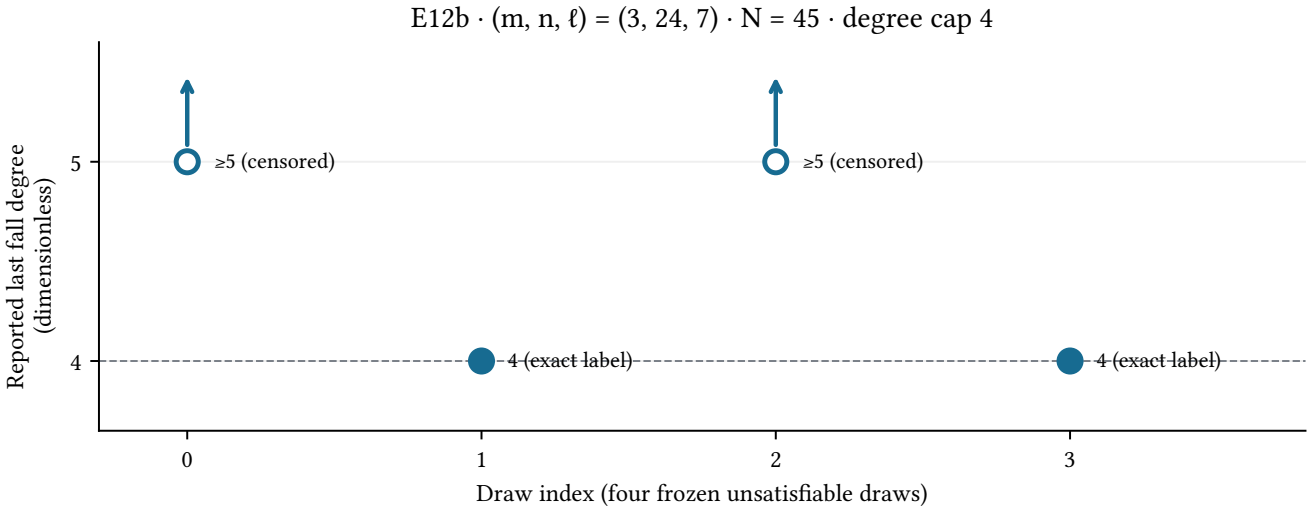
E12b: completed Spot continuation

Observed records and frozen scoring · 28 October 2026

This report records solver outputs and applies a frozen arithmetic rule. It is not an independent closure replay, hypothesis promotion, or proof of a general fall-degree bound. Four draws completed for $(m, n, \ell) = (3, 24, 7)$, $N = 45$, 48 equations, seed 20261012 and degree cap 4.



Four recorded labels; upward arrows denote censoring, not exact degree 5. Job: 20261028-031317-fd-e12b-spot-20261028-r3-24148b. Raw SHA-256: 055811b722f441ab681172a605f9a540b174734dbb4f23ff9f03b931d948c81b.

Draw	Recorded label	Degree-4 dimension / columns	Rounds
0	≥5	157926 / 164221	9
1	4	164221 / 164221	8
2	≥5	158254 / 164221	9
3	4	164221 / 164221	7

Frozen reading: “mixed”

Amendments 12/15 score censored rows as cap plus one and take the second-smallest of four values. Sorting $[5, 4, 5, 4]$ gives $[4, 4, 5, 5]$, so the lower median is 4. Two rows score at least 5; the rule requires at least three. The other three carried-forward r2 cells each retain four values of 4. Earlier partial draw 0 is an identity reference, not an additional sample. No significance test is claimed.

Execution and checks

Four requested draws completed; none failed or were interrupted in this attempt. All record zero solutions. The m7i.4xlarge Spot host had 16 vCPU and 64 GiB in us-east-1. Job status records exit 0; the completed instance terminated.

Gate logs record three counter tests, 24 gate-1 agreements, 16 gate-2/3 matching rows, and draw-0 equality to r2 after removing only seconds fields. A separate record audit checked these against frozen references, without executing a solver. Diagnostic solver intervals sum to 1203.0 seconds, excluding build and gates. The whole-command rusage records 24916.64 user CPU seconds, 505.06 system CPU seconds and 30:57.68 elapsed, including build and gates. Timing is not scored or compared.

Limits and provenance

The two censored degrees remain unknown above the cap. Record consistency does not independently establish unsatisfiability, saturation, or engine correctness. No new Lean verification is claimed; the prior 282-row table excludes these rows. General first-fall assumptions and asymptotic last-fall bounds remain unresolved.

Source commit: 371035216942660c4615dfbb88d85a14afaf5c16.
Frozen protocol: 020f03193c46d975935a065c966c81941cbdbf6d, Amendments 12/15.
Archive SHA-256: b06c3745486ecf1e2e5fce7818e1495de06cb4bcfaa09e644098db379d235d9d.
See accompanying README.md, provenance.json and verification.json for S3 location, exact commands, identities, gates and limitations.
verify_records.py reproduces archive consistency and scoring; plot.py regenerates this graph from the raw rows.